How well do X-ray cavities measure jet energetics?

Abstract

Keywords

Introduction

Numerical jet experiments embedded in realistic cosmological environments provide a controlled link between source dynamics and synthetic observables (Yates-Jones *et al.* 2023). Synthetic X-ray observations further show that cavity enthalpy can be recovered reasonably well while source age and viewing angle remain important sources of uncertainty (Mendygral *et al.* 2011).

Methods

Simulation design

We compare four jet simulations crossing cluster and group atmospheres with high and low jet power (C-HI, C-LO, G-HI and G-LO). The imposed bipolar powers are 2×10^{45} and 2×10^{43} erg s^{-1}, respectively. The analysis uses outputs near 10, 20, 30, 40 and 50 Myr, with actual output times retained in all energy and power calculations. G-HI at 50 Myr is excluded because the source reaches the simulation boundary, leaving 19 case-age combinations. These four simulations are factorial cases, not replicated samples of an astrophysical population.

Synthetic X-ray and radio products

Cavity geometry, age, and energetics

The primary native cocoon comprises cells with tracer tr1 $> 10^{-3}$, evaluated after promotion of the stored tracer to double precision. Each selected cell contributes its full volume, mass and energy. Ambient cells satisfy the exact complement, tr1 $\leq 10^{-3}$. Fractional tracer weighting and binary thresholds of 5×10^{-3} and 10^{-2} provide sensitivity comparisons. The stored single-precision tracer can differ slightly from the nominal threshold; the largest contribution of cells at its stored representation is less than 2.6×10^{-7} of native enthalpy on the selected surface.

Native enthalpy is $H_{\rm native}=\alpha\int_{\rm cocoon}p\,dV$, with $\alpha=2.5$ for the primary non-relativistic convention and $\alpha=4$ as a literature comparator. Thermal, kinetic and gravitational energies form a separate additive ledger. For each component we subtract the initial field in the same cells selected by the current cocoon mask, and do likewise for its ambient complement. These regional differences sum to the retained whole-domain energy change. They describe changes relative to the initial conditions in a current spatial region; they do not follow a material parcel or the evolving cocoon boundary. The initial snapshots already contain a nozzle region. Absolute current lobe energy therefore differs from the regional change, and subtracting the entire initial domain energy from the ambient allocation is not equivalent to subtracting only the initial ambient-region energy.

We identify shock-processed ambient gas using entropy excess connected to a compressed axial-front seed, with environment-matched no-jet noise calibration. The calculation retains the accepted supersonic-front, seed, extent, boundary, minimum-cell and bilateral validity gates. Selected cells contribute their full current-minus-initial energy, and are a subset of the ambient region. Shock energy must therefore not be added again to the full ambient energy change. The calibration is not a native-resolution cellwise no-jet subtraction. Boundary energy flux and radial advection are unavailable; agreement of the numerical partition identities does not establish a closed physical energy budget.

X-ray cavity measurements are evaluated independently of tracer-based membership. The primary comparison uses geometric depth, pressure averaged over the inferred ellipsoid, and the same enthalpy convention as the native estimate. We evaluate viewing angles of 0, 45 and 90 degrees and require a valid bilateral cavity topology. Invalid views remain unresolved and do not enter quantitative recovery summaries.

Statistical comparisons

We report within-case ranges and medians across the accepted ages and views. These are descriptive summaries of repeated measurements, without population-level significance tests or an assumption that ages and views are independent. A factor of two provides a common descriptive recovery band. Spectral breaks above the fitted frequency range yield upper limits on age and lower limits on inferred power; these limits remain separate from exact estimates. Sound-crossing, buoyant-rise and refill clocks use the

retained initial-atmosphere sound speed and gravity with the observed component geometry. The mean-pressure spectral clock retains its previously defined fractional-tracer global magnetic-field estimate; it does not use the X-ray pressure or the revised binary-cocoon pressure. The emissivity-weighted field supplies its separate sensitivity comparator.

Results

Does cavity enthalpy capture the injected jet energy?

With the true age known, binary-cocoon enthalpy captures a substantial but case-dependent fraction of injected energy (Figure 1). Native $2.5pV/E_{\rm inj}$ spans 0.525–0.559 in C-HI, 0.468–0.498 in G-HI and 0.615–0.785 in G-LO. C-LO spans 0.631–1.108. Switching to $4pV$ multiplies every value by 1.6; it changes the normalization but does not supply a universal correction across the four cases.

Enthalpy also differs from the actual energy change in the current cocoon region. The latter spans 0.132–0.426 of injected energy in C-HI, 0.425–0.571 in G-HI and 0.013–0.298 in G-LO. In C-LO it ranges from −0.160 to 0.137. Negative gravitational changes relative to the initial fields contribute to this separation, and a negative net regional value does not imply negative absolute energy or a Lagrangian loss from the jet. Thermal pressure alone cannot represent all terms in this accounting.

The selected shock-processed ambient region contains energy changes of 0.666–0.765, 0.505–0.655 and 0.580–0.854 times the injected energy in C-HI, G-HI and G-LO, respectively. C-LO supplies one accepted estimate, 0.223 at 10 Myr; its four later ages fail the bilateral shock gate. This selected energy includes thermal, kinetic and gravitational changes and is not entirely irreversible heat. The estimated irreversible-heating term spans 0.177–0.229, 0.257–0.283 and 0.120–0.263 of injected energy in the three fully resolved series.

The numerical cocoon/ambient decompositions agree with the independently summed retained energy change to the prescribed tolerance of $10^{-9}E_{\rm inj}$. Physical closure remains unresolved: the retained whole-domain change exceeds injected energy, particularly in C-LO, where the ratio is approximately 6–24. That case's background-dominated ambient budget cannot be interpreted as jet coupling efficiency. The shock measurements support substantial energy outside the native cocoon, while neither their restricted masks nor the whole-domain ledger establish the fraction of jet power irreversibly coupled to the atmosphere.

How accurately do X-ray observations recover cavity enthalpy?

Of the 57 selected case-age-view combinations, 37 pass the bilateral X-ray topology criterion and 20 remain unresolved. Within the valid surface, pressure recovery is within a factor of two for 36 of 37 views, compared with four for volume and five for enthalpy (Figure 2). Thus the dominant measurement discrepancy lies in the recovered cavity extent and topology, rather than in the inferred ambient pressure. This conclusion is conditional on successful cavity identification; excluded views are not assigned small but finite cavity energies.

Median X-ray/native enthalpy ratios are 0.0222 in C-HI, 0.494 in C-LO, 0.00603 in G-HI and 0.0929 in G-LO. The low-power cluster case is the closest median recovery, and the high-power cases show the largest deficits. These values use the binary cocoon required for the native comparison. Fractional tracer weighting gives a C-LO median ratio of 6.76, so its apparent over-recovery under that alternative convention disappears when the full binary region defines native truth. The observer measurements themselves are unchanged. This sensitivity changes the interpretation of C-LO but preserves the broader evidence for geometry-dependent recovery.

How do uncertain ages affect inferred jet power?

For constant bipolar power, the three contributions to an inferred-power error can be written

$$\frac{Q_{\rm cav}}{Q_{\rm true}} = \frac{H_{\rm native}}{E_{\rm inj}} \frac{H_X}{H_{\rm native}} \frac{t_{\rm true}}{t_{\rm est}}.$$

The identity separates the native enthalpy proxy, observational recovery and age estimate, using the same bipolar region and enthalpy convention throughout. When ages are estimated separately for the two sides, we retain the summed power $\sum_s H_{X,s}/t_s$ and define $t_{\rm est} = H_X/(\sum_s H_{X,s}/t_s)$ for this identity. This effective age is an enthalpy-weighted harmonic mean, distinct from the arithmetic mean used to summarise the two side ages. It does not assume that the factors are statistically independent. Changing the native mask changes the first two factors reciprocally, leaving their product unchanged for a fixed X-ray measurement.

Using X-ray $2.5pV$ and the true age, median inferred-to-known power ratios are 0.0122, 0.514, 0.00292 and 0.0725 for C-HI, C-LO, G-HI and G-LO, respectively. The high-power deficits are therefore already present before any clock error is introduced. Sound-crossing, buoyant-rise, refill, shock and spectral clocks add method-dependent shifts (Figure 3). Among exact mean-pressure spectral estimates, the corresponding medians are 0.0181 for C-HI and 0.856 for C-LO. However, 20 of 37 mean-pressure spectral rows are censored; both group cases supply only power lower limits for this clock. These limits cannot establish a finite correction to their energy deficit. Emissivity-weighted spectral ages remain a sensitivity comparison.

Discussion

The four simulations separate two limitations of cavity calorimetry. Even an exactly measured native enthalpy is a conditional proxy for jet energy because the pressure integral omits kinetic and gravitational changes and does not isolate energy deposited outside the cocoon. The independent X-ray measurement then introduces a larger deficit in the high-power cases through cavity geometry and topology. An age correction acts on that already biased energy estimate.

The binary native definition is consequential for the low-power cluster case. It converts apparent X-ray over-recovery under fractional tracer weighting into a median underestimate. This illustrates why the definition of native truth must accompany a recovery claim. It does not justify using tracer to repair an observed cavity or applying a single empirical correction across environments and powers.

Shock-mask sensitivity also limits the precision of energy-deposition claims. Halving and doubling the shock-mask thresholds leaves the G-HI shock energy almost unchanged, but produces appreciable variation in G-LO and in C-LO's sole resolved point. Together with the absent boundary flux and cellwise no-jet counterfactual, this warrants interpreting the shock integral as a selected-region diagnostic. The present comparison establishes the structure of the bias within four deterministic cases; a population calibration or causal coupling efficiency requires additional evidence.

Conclusions

Data availability

Code availability

Author contributions

Financial support

Conflicts of interest

None.

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AI-use declaration

References

Mendygral, P. J., O'Neill, S. M., & Jones, T. W. (2011). Synthetic observations of simulated active galactic nucleus jets: X-ray cavities. *The Astrophysical Journal*, **730**(2), 100.

Yates-Jones, P. M., Shabala, S. S., Power, C., … Stewart, G. S. C. (2023). CosmoDRAGoN simulations - I. Dynamics and observable signatures of radio jets in cosmological environments. *Publications of the Astronomical Society of Australia*, **40**, e014.

Figure Captions

Fig. 1. Native enthalpy and selected-region energy changes relative to the known injected bipolar energy. Each panel shows one simulation at its accepted actual ages. The native cocoon uses full-cell tracer membership above 10^{-3}. Blue and orange curves show current $2.5pV$ and $4pV$, respectively; these are proxy energies and are not additive terms in the ledger. Green curves show thermal, kinetic and gravitational changes relative to the initial fields in the cells comprising the current cocoon. Purple curves show the corresponding change in the selected shock-processed ambient region. The shock region is a subset of ambient material. C-LO has only one quantitative shock point; later failures are left unresolved. G-HI at 50 Myr is excluded. The dashed line marks injected energy, without asserting physical closure of the retained domain.

Fig. 2. Conditional X-ray recovery of binary native volume, pressure and enthalpy. Points are individual accepted case-age-view combinations; horizontal coloured bars and labels give within-case medians. The pressure ratio compares the effective ellipsoid-averaged X-ray ambient pressure with the volume-averaged native cocoon pressure. The primary observer convention is geometric depth and $2.5pV$. Grey bands mark the descriptive factor-of-two interval around unity. The panels contain the same 37 valid views (11 C-HI, 10 C-LO, six G-HI and 10 G-LO); 20 topology failures are excluded without imputation. Ages and views are repeated measurements, not independent simulation replicates.

Fig. 3. Clock estimates and cavity-power recovery using the accepted X-ray-deprojected enthalpies. Filled circles show medians of exact estimates within each case and method. Open downward and upward triangles show median reported bounds for age upper limits and power lower limits, respectively; these summarise the censored subset and are not estimates of its unknown median. The age panel uses the arithmetic mean where two side ages are available. Power for side-based clocks is the sum of each side's enthalpy divided by its own age, whereas true and spectral clocks use a common age. The effective age in the power-factorisation identity therefore differs from the arithmetic mean. Missing shock ages remain excluded. Sound, buoyant and refill methods retain initial-atmosphere sound speed and gravity. The mean-pressure spectral field retains its fractional-tracer convention, with an emissivity-weighted alternative shown separately. Spectral limits remain limits, including all mean-pressure spectral points in the two group cases.

Fig. 1.

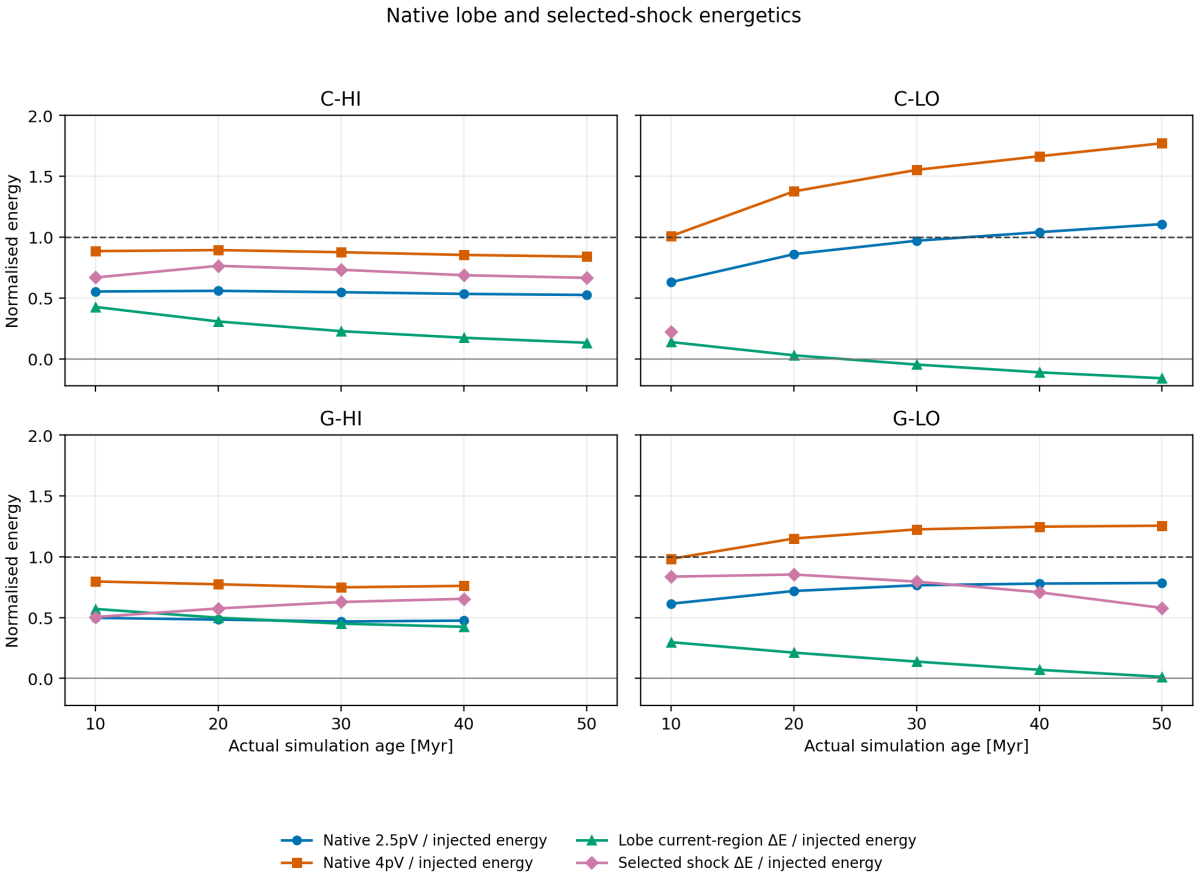

Native lobe and selected-shock energetics
C-HI
C-LO
G-HI
G-LO
Normalised energy
Actual simulation age [Myr]
Native 2.5pV / injected energy
Native 4pV / injected energy
Lobe current-region ΔE / injected energy
Selected shock ΔE / injected energy

Fig. 2.

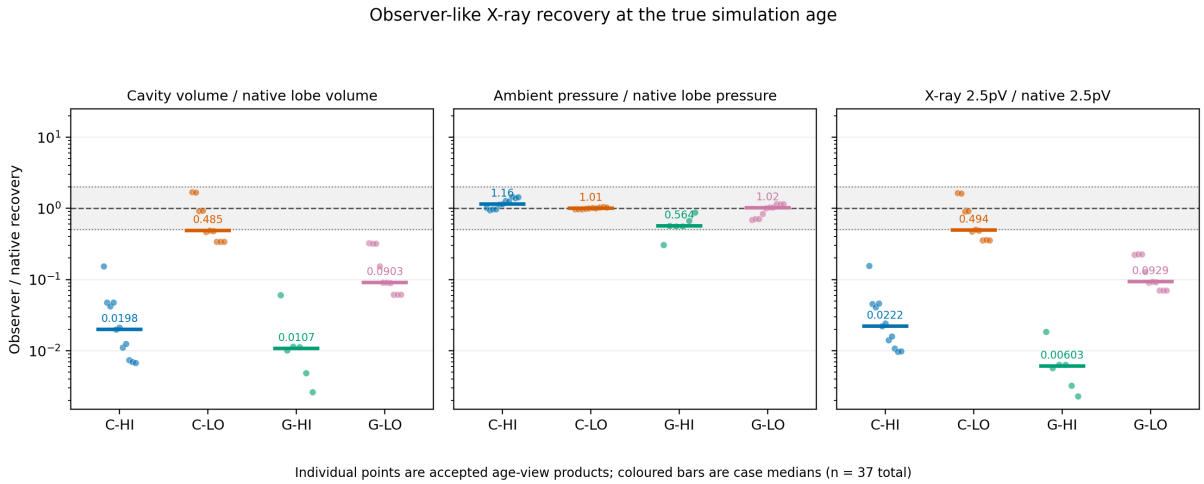
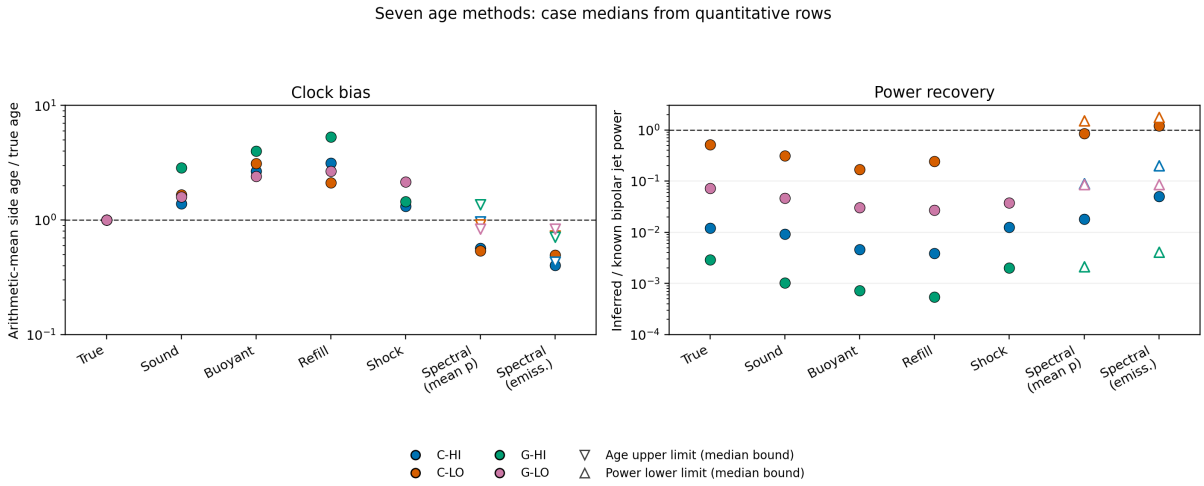

Observer-like X-ray recovery at the true simulation age
Cavity volume / native lobe volume
Ambient pressure / native lobe pressure
X-ray 2.5pV / native 2.5pV
Observer / native recovery
10^{1}
10^{0}
10^{-1}
10^{-2}
0.0198
0.485
0.0107
0.0903
1.16
1.01
0.564
1.02
0.0222
0.494
0.00603
0.0929
C-HI
C-LO
G-HI
G-LO
Individual points are accepted age-view products; coloured bars are case medians (n = 37 total)

Fig. 3.


Seven age methods: case medians from quantitative rows
Clock bias
Arithmetic-mean side age / true age
Power recovery
Inferred / known bipolar jet power
True
Sound
Buoyant
Refill
Shock
Spectral (mean p)
Spectral (emiss.)
C-HI
C-LO
G-HI
G-LO
Age upper limit (median bound)
Power lower limit (median bound)